Phase 2 Pre-Registration: Calibrated Reliability Interface vs Raw Confidence

Date: 2026-06-19 (revised 2026-06-23 — leakage controls, multiplicity correction, and honest cutoff/CA-lDDT framing added before deposit; not yet registered on OSF) Status: PRE-REGISTRATION for the robust redo. The exploratory Phase-2-v1 run (80 targets, Sonnet+Opus, RESULTS.md "Phase 2") is treated as pilot/exploratory; its offline-gate PASS was engineered post-hoc and its calibration was fit in-distribution. This document fixes the analysis BEFORE the confirmatory run so the result is not a garden-of-forking-paths.

What v1 established, stated honestly: the v1 LLM routing contrasts look invariant to the lDDT cutoff (0.5-0.9; `results/phase2_preflight/cutoff_robustness.json`), but that invariance is largely a SATURATED-LABEL artifact – at lDDT<=0.8 almost every v1 target is "correct" (base rate ~95%), so re-scoring the same fixed actions against relabeled truth barely moves (interface-raw is exactly 0.0 for Sonnet and -0.25 for Opus at every cutoff). It is therefore NOT independent evidence that the 0.9 cutoff is harmless. The honest reading is the opposite: at the standard lDDT>=0.7 cutoff the offline gate returns AUROC ~0.60 – the same near-noise verdict that retired the Phase 1 single-cell arm – and v1's gate PASS exists only because the cutoff was moved to 0.9 post-hoc AND the calibration was refit in-distribution. The confirmatory redo removes BOTH crutches (pre-set 0.7 cutoff, held-out calibration) and instead earns routing stakes from a genuinely lower base rate.

Question (unchanged North Star)

Does converting a validated specialist-confidence signal into a calibrated reliability *interface* improve an LLM's cost-aware trust routing OVER simply showing the raw confidence – and is that effect model-general?

Pre-registered hypotheses (directional, from v1)

- H1 (expected SUPPORT): any confidence cue beats the no-signal baseline.
- H2 (expected NULL / model-dependent): the calibrated card does NOT beat raw pLDDT; specifically ~0 for Sonnet and NEGATIVE for a risk-averse model (Opus). This is the primary, confirmatory test – registered as a likely null.
- H3 (expected SUPPORT): the benefit is informational, not directive – the no-recommendation arm equals the full card (A4).
- H4 (expected SUPPORT): an inverted card degrades routing vs the calibrated card.

A confirmatory NULL on H2 is the headline; we are testing it honestly, not hunting for a positive.

Design (fixes the v1 caveats)

- Substrate: protein structure (Boltz-2 / pLDDT), post-training-cutoff targets.
- Targets: powered N with a LOWER base rate so routing has stakes – include harder targets (e.g., recent CASP-hard / low-homology / larger complexes) so the specialist is wrong often enough; pre-specify N per regime by a power calculation for the H2 effect size, not a default.
- Correctness metric: lDDT VALIDATED against OpenStructure (monomers) + **DockQ / interface-lDDT (complexes)**, not the home-grown CA-lDDT. (v1's "all-chain CA-lDDT" for complexes is intra-chain-dominated – most sub-15A CA pairs are within a chain, so

it mostly re-measured per-chain fold quality, not docking; DockQ is the correct interface axis and replaces it.) Cutoff PRE-SET (lDDT>=0.7, the standard "modelable" threshold; DockQ>=0.23 "acceptable" for complexes) with a full sensitivity sweep reported.
- Calibration: fit risk->P(wrong) on a HELD-OUT split (train calibration on a disjoint target set; never in-distribution LOO over the evaluated set).
- Arms (>=6): no_signal, raw_plddt_shown, calibrated_risk_shown_no_recommendation, calibrated_interface_shown, inverted_reliability_interface_control, and a NEW competing-cue arm (e.g., correct confidence paired with a misleading model-name/provenance cue) so "the LLM uses the signal" is not forced by the packet containing only confidence.
- Models: >=3 (e.g., Sonnet, Opus, Haiku and/or a non-Anthropic model), with repeated sampling (>=3 seeds/temperature draws) so cross-model differences get their own CIs.
- Reward: net = correct - lambda*assays; lambda sweep {0.2,0.5,0.8}; a REAL template/homology baseline for default_baseline (revive that action), not the current default-False placeholder.

Leakage controls (pre-specified)

A post-cutoff release date is necessary but NOT sufficient; v1 controlled leakage by date alone. The confirmatory run pre-registers three additional, reported controls:

- Training-cutoff date: use Boltz-2's ACTUAL structural-data cutoff, not the most generous quotable date. The curation code itself notes structural/complex data flowed in "through early 2025" (`phase2_curate_pdb.py`), so targets are curated with `released_after = 2025-07-01` (not 2023-06), and the exact date + source is reported on every public card. No "2023-06 leakage-safe" framing.
- Sequence-identity dedup: for every target chain, compute max sequence identity to any pre-cutoff PDB chain (MMseqs2 / `phmmer`); PRE-EXCLUDE any target with >30% identity to a pre-cutoff chain (near-duplicate / re-determination / point-mutant / alternate crystal form), and report the threshold and the excluded count. A structural-redundancy check (Foldseek TM-score vs pre-cutoff PDB) is reported as a secondary filter.
- MSA-depth as a reported covariate: Boltz-2 is MSA-conditioned, so a post-cutoff target whose family has pre-cutoff homologs still gets a deep MSA and a justified high pLDDT – a leakage channel the date cannot close. Log Neff / MSA depth per target; the primary analysis covaries/stratifies on MSA depth, and a low-MSA (shallow-alignment) subgroup is reported separately as the hardest, least-leaky slice.

Analysis plan (pre-set)

- Primary contrast: calibrated_interface_shown vs raw_plddt_shown, per model, target-bootstrap 95% CI (seed 13, 1000 draws). Decision: SUPPORT only if the CI excludes 0 in the positive direction; otherwise report the null / negative.
- Multiplicity: H2 (interface vs raw) is the SINGLE confirmatory primary; everything else is secondary/exploratory. Across the registered secondary family (arms x models x lambda x cutoff) report Holm-Bonferroni-adjusted intervals, and flag any unregistered comparison as exploratory. The headline rests on the primary only.
- Secondary: each arm vs no_signal; A4 (card vs no_recommendation); inverted vs calibrated; monomer/complex split; lambda sweep; MSA-depth subgroup.
- Disentangle the regime LABEL from the risk VALUE (regress verify on risk within regime) so "regime-appropriate caution" is not a label artifact.

Success / failure framing (both publishable)

- If H2 SUPPORTED: calibrated interfaces help over raw confidence – a presentation-layer win.
- If H2 NULL/NEGATIVE (expected): "prompt-visible reliability interfaces are not a robust lever over raw calibrated confidence and can backfire by model; presentation is insufficient – enforcement (tools/MCP/post-training) is the next lever." This motivates Phase 4.

Stop rules

- Do not re-pick the cutoff or calibration after seeing results.
- Do not add modalities/SFMs before this confirmatory run is scored.
- Freeze and report the leakage-exclusion list (sequence-identity / structural / date filters) and the per-target MSA-depth covariate BEFORE scoring any LLM arm.
- Report the template-baseline and lDDT/DockQ-validation status explicitly; no claim rests on the home-grown CA-lDDT.